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كلية الحاسبات والنكاء الاصطناعي

Project Name: ...Room Escape(Maze) Game...

Doc	3D	Keyboard and mouse	Collision	Camera	Texture	Lighting

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Room Escape

Maze Escape is a 3D computer graphics game developed using OpenGL and GLUT in C++. The project focuses on implementing important computer graphics concepts such as lighting, textures, camera movement, collision detection, and interactive gameplay mechanics.

The player is placed inside a 3D maze and must explore the environment to find a golden key before the timer runs out. After collecting the key, the player can open the exit door and move to the next level. The game contains two levels with different maze layouts and challenges.

The project uses textured walls, floors, and doors to create a more realistic environment. Lighting techniques including ambient, diffuse, and specular lighting are applied to improve the visual appearance of the scene. The player can move inside the maze using keyboard controls while the camera rotates using mouse movement.

Main Features

- 3D maze environment
- Two playable levels
- Textured walls, floor, and doors
- Real-time lighting system
- Collision detection with walls
- Collectable key system
- Timer and game over system
- Win and lose screens
- Mouse-controlled camera rotation
- Full screen mode support



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How to Play

Objective

The main goal of the game is to escape the maze before the timer reaches zero.

To complete the game:

1. Explore the maze
2. Find the golden key
3. Reach the exit door
4. Complete both levels before time runs out

Controls

KEY	FUNCTION
W	Move Forward
S	Move Backward
A	Move Left
D	Move Right
MOUSE MOVEMENT	Rotate Camera
F1	Toggle Full Screen
R	Restart Game
Q / ESC	Exit Game

Gameplay Instructions

Step 1 — Start the Game

When the game begins, the player appears inside the maze and a short instruction message is displayed.

Step 2 — Explore the Maze

Use the keyboard controls to move through the maze while avoiding walls.



Step 3 — Find the Golden Key

The player must collect the floating golden key before attempting to escape.

- Without the key, the door remains locked.
- Approaching the door without the key displays a warning message.

Step 4 — Open the Door

After collecting the key:

- The door opens automatically.
- The player can proceed to the next level or finish the game.

Step 5 — Complete Level 2

The second level contains a different maze layout and another key.

The player wins after:

- Collecting the second key
- Reaching the final exit door before time runs out

Win Condition

The player wins when:

- Both levels are completed
- The final exit door is reached
- Time has not expired

A winning screen is displayed after successful completion.

Lose Condition

The player loses when:

- The timer reaches zero before escaping the maze

A game over screen is displayed.